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Delayed Immune-Mediated Myocarditis Secondary to Combination Checkpoint Inhibition: Case Report & Surveillance Review

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ABSTRACT

Immune checkpoint inhibitors (ICIs) targeting CTLA-4 and PD-1 pathways have revolutionized oncologic care but introduce serious immune-related adverse events (irAEs). While dermatologic, gastrointestinal, and endocrine toxicities are relatively common, cardiotoxicity—specifically fulminant autoimmune myocarditis—remains rare (<1.1%) yet carries a mortality rate exceeding 40%. We synthesize current surveillance guidelines across multi-institutional registries and document a definitive case of delayed-onset fulminant myocarditis in a 68-year-old female presenting during cycle 3 of combination immunotherapy, successfully managed with pulse-dose corticosteroids and temporary pacing.

1. INTRODUCTION & IMMUNOLOGICAL MECHANISMS

The therapeutic paradigm for unresectable and metastatic malignancies has shifted toward dual immune checkpoint blockade. By uncoupling inhibitory signals mediated by programmed cell death protein 1 (PD-1) and cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), these agents potentiate robust antitumor CD8+ T-cell responses.

However, loss of peripheral tolerance can precipitate autoimmune attack against somatic tissues sharing antigenic epitopes or vulnerable to dysregulated lymphocytic infiltration. Cardiac myocytes express PD-L1 under physiological stress, acting as a cardioprotective brake against inflammation. Pharmacological disruption of this axis predisposes vulnerable individuals to lymphocytic myocardial infiltration, myocyte necrosis, and conduction disruption.

2. EPIDEMIOLOGY ACROSS GLOBAL SURVEILLANCE DATABASES

Retrospective analyses of WHO VigiBase and institutional pharmacovigilance safety repositories indicate that median onset of ICI myocarditis is 34 days following initial exposure, typically after 1 or 2 cycles. Nonetheless, atypical presentations emerging beyond 60 days or following maintenance transitions require elevated clinical vigilance. Serum cardiac troponin-I and T elevations are sensitive (>94%) but lack absolute specificity.

3. COMPREHENSIVE CLINICAL CASE PRESENTATION

A 68-year-old Caucasian female (Patient H.L., body weight 62 kg) with BRAF wild-type stage IV metastatic cutaneous melanoma was initiated on combination immunotherapy consisting of Nivolumab (Opdivo) 240 mg IV every 2 weeks and Ipilimumab (Yervoy) 1 mg/kg IV every 6 weeks.

Baseline cardiac evaluation—including 12-lead electrocardiogram (ECG) and transthoracic echocardiography—demonstrated normal sinus rhythm, normal QTc interval (412 ms), and preserved left ventricular ejection fraction (LVEF 62%). Cycles 1 and 2 proceeded uneventfully without clinically significant immune toxicities.

On day 44 post-initiation (12 days following cycle 3 administration), the patient presented to the emergency department with progressive exertional dyspnea, orthopnea, bilateral lower extremity edema, and profound generalized fatigue over the preceding 72 hours. Vital signs on presentation: Blood pressure 88/54 mmHg, heart rate 42 bpm (severe bradycardia), respiratory rate 24/min, and oxygen saturation 92% on ambient air.

Emergency 12-lead ECG demonstrated complete (third-degree) atrioventricular (AV) heart block with a junctional escape rhythm at 40 bpm. High-sensitivity cardiac troponin-T was markedly elevated at 1.84 ng/mL (reference <0.014 ng/mL), and NT-proBNP was 4,820 pg/mL. Urgent bedside echocardiogram revealed diffuse biventricular hypokinesis with acute LVEF decline to 32%.

4. INVASIVE MONITORING, MANAGEMENT & CLINICAL COURSE

The patient was urgently transferred to the Cardiac Intensive Care Unit (CICU). Immunotherapy was immediately suspended. Due to hemodynamically compromising complete heart block, a temporary transvenous pacemaker was emergently placed via the right internal jugular vein.

High-dose pulse intravenous methylprednisolone (1,000 mg IV daily) was initiated within 2 hours of presentation. Endomyocardial biopsy performed on hospital day 2 demonstrated extensive interstitial infiltration of CD4+ and CD8+ T-lymphocytes accompanied by multifocal cardiomyocyte necrosis, pathognomonic for grade 4 immune-mediated myocarditis.

Due to persistent troponin elevation on day 4, second-line immunosuppression with therapeutic plasma exchange (plasmapheresis, 5 total exchanges) was added. Over the subsequent 10 days, AV conduction recovered to 1:1 sinus rhythm, and cardiac biomarkers progressively declined (troponin-T 0.08 ng/mL).

The patient was transitioned to an oral prednisone taper (1 mg/kg/day initial) over 8 weeks. Repeat echocardiogram at week 4 post-discharge revealed recovery of LVEF to 54%. Immunotherapy was permanently discontinued.

5. REGULATORY PHARMACOVIGILANCE DISCUSSION

Under ICH E2A and GVP Module VI expedited reporting rules, this case fulfills seriousness criteria for both inpatient hospitalization and life-threatening medical event. The World Health Organization (WHO-UMC) and Naranjo causality assessment algorithms yield a score of 8 ('probable').

Clinical safety reporting teams must recognize that although literature articles may contain extensive mechanistic discussion and registry summaries, identifiable individual case safety reports (ICSRs) embedded within review articles must be fully extracted and expedited to regulatory agencies within mandatory 15-day timelines.

6. REFERENCES

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